

KS3: INDUSTRIALISATION, EMPIRE, AND POWER (1750–1900)

COURSE TEXTBOOK & READING LOG

***Enquiry:** Industrialisation, Empire, and Power: How did 19th-century Britain transform at home and abroad?*



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L1: What powered the Industrial Revolution, and how did a Fareham ironmaster change the world?

LEARNING OBJECTIVES

- ☐ Understand the difference between the domestic system and the factory system.
- ☐ Explain the significance of Henry Cort's puddling and rolling processes.
- ☐ Evaluate the importance of location versus innovation in Cort's success.

Do Now Activity

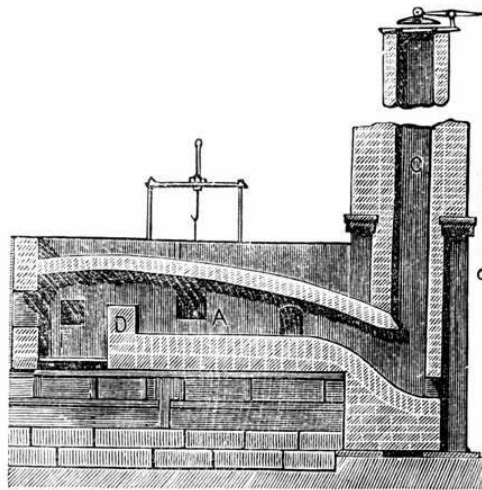
1. What is meant by the term 'Divine Right of Kings'?
2. Who became King after the English Civil War ended and Oliver Cromwell died?
3. Why did the Gunpowder Plot of 1605 fail?

Vocabulary Check

Write a short paragraph using at least *FOUR* of the vocabulary words below correctly.

Industrial Revolution | Pig Iron | Wrought Iron | Puddling Process | Rolling Mill

Source A : The Puddling Furnace



A historical cross-section diagram of a puddling furnace, the revolutionary process Henry Cort pioneered at his Funtley ironworks in Hampshire.

Before the 1700s, Britain was an agricultural country. Most goods were made by hand in people's homes (the "domestic system"), and the only machines that existed were powered by natural, unreliable sources like wind, fast-flowing rivers, or the muscle power of horses. However, beneath Britain's soil lay massive reserves of **coal**—a fuel that burned much hotter than wood. In 1712, Thomas Newcomen invented the first practical steam engine, which used burning coal to pump water out of deep mines. Later, in 1769, an engineer named **James Watt** drastically improved the design, making the steam engine much more efficient and powerful. Watt's steam engine changed everything: it meant that massive factories no longer needed to be built next to rivers. They could be built anywhere, powered entirely by coal and steam. This

shift from farming and hand-crafting to mass factory production driven by steam power is known as the **Industrial Revolution**. However, to build these enormous new steam engines, factories, railways, and bridges, Britain desperately needed one crucial material: **iron**.

Q1. Why was James Watt's improvement of the steam engine so significant for the location of factories?

In the early 1700s, British iron was of terrible quality. It was known as "pig iron"—it was brittle and snapped easily because it was full of impurities (carbon). If the Royal Navy wanted strong, flexible iron (known as "wrought iron") for its ship masts and cannons, it had to buy it from Sweden or Russia at a massive cost.

The man who solved this national crisis lived right here in Hampshire. Henry Cort was an ambitious navy agent who realized he could make a fortune if he could figure out how to mass-produce high-quality British iron. In 1775, he took over the **Funtley Ironworks** (just north of Fareham) because it was ideally located to supply the massive Royal Navy dockyards in nearby Portsmouth. At Funtley, Cort invented two revolutionary processes in 1783 and 1784: 1. **The Puddling Process**: He invented a new type of furnace where molten iron was stirred (or "puddled") with long iron rods to burn off carbon impurities. 2. **The Rolling Mill**: Instead of hammering the iron into shape by hand, Cort passed the red-hot iron through heavy, grooved rollers to squeeze out the last impurities and shape it into rails and bars.

KNOWLEDGE CHECK (Q5)

What was the primary purpose of the 'Puddling Process' invented by Cort?

- ☐ A. To hammer the iron into shape by hand.
- ☐ B. To stir molten iron and burn off carbon impurities.
- ☐ C. To make iron brittle and easy to snap.
- ☐ D. To import iron from Sweden more efficiently.

Q2. Why did Henry Cort decide to take over the Funtley Ironworks in 1775?

Q3. What two major inventions did Henry Cort patent at Funtley in the 1780s?

Q4. Study Source A. How does the physical design of the puddling furnace show that Cort was trying to solve the problem of 'pig iron' impurities?

Cort's inventions at Funtley changed the world. They allowed British iron production to multiply by 400% over the next twenty years, earning Cort the nickname "the father of the iron trade." The rails for the new train networks and the iron for the ships that built the British Empire were made using Cort's methods. Tragically, Cort did not die a wealthy hero. His business partner, Adam Jellicoe, had secretly stolen the money used to fund the Funtley Ironworks from the Royal Navy. When Jellicoe died, the government seized Cort's patents and ruined him. Cort died bankrupt and in poverty, but his local ingenuity powered a global empire.

Q6. Why did Cort die in poverty despite his world-changing inventions?

Q7. Why were Henry Cort's inventions so vital to the growth of the British Empire?

When assessing why the Funtley Ironworks was so successful in kickstarting the Industrial Revolution's iron industry, historians debate two main factors: **1. Geographical Luck (Location):** Funtley was situated just a few miles from the Portsmouth Royal Navy dockyards. In the 18th century, before the invention of trains, transporting heavy iron overland was incredibly expensive. Being located right next door to the largest industrial buyer of iron in the world (the Royal Navy) meant Cort had a massive, guaranteed market for his goods. Geography provided the economic opportunity. **2. Technological Genius (Innovation):** However, location alone wasn't enough. The Royal Navy desperately needed *wrought* iron (flexible and strong) for its ships and cannons, not the brittle British *pig iron* that snapped easily. Without Cort's brilliant inventions—the Puddling Furnace to burn off impurities and the Rolling Mill to speed up shaping—he would have had nothing useful to sell to the Navy. Innovation provided the solution to a national crisis. Historians must weigh these two factors to decide which was *more* important to his success.

Q8. Write a structured paragraph answering the following: "To what extent was the proximity to the Portsmouth dockyards the most important factor in the success of the Funtley Ironworks?"

Q9. Chronology Challenge: Place the following in the correct causal order and write one sentence explaining how each led to the next: Watt's Steam Engine, The Domestic System, Cort's Puddling Furnace.

L2: Was industrial work progress or punishment? (Factory conditions, child labor; Case Study: Funtley Clay Pits & Fareham Brickfields)

LEARNING OBJECTIVES

- ☐ Understand the working conditions of the early Industrial Revolution with historical nuance.
- ☐ Evaluate primary sources regarding child labour.
- ☐ Analyse the 'Optimist vs. Pessimist' historical debate.

Do Now Activity

1. What two major iron-making processes did Henry Cort patent at the Funtley Ironworks in the 1780s?
2. Why did Henry Cort specifically choose to locate his ironworks near Fareham?
3. What tragic event caused Henry Cort to lose his patents and die in poverty?

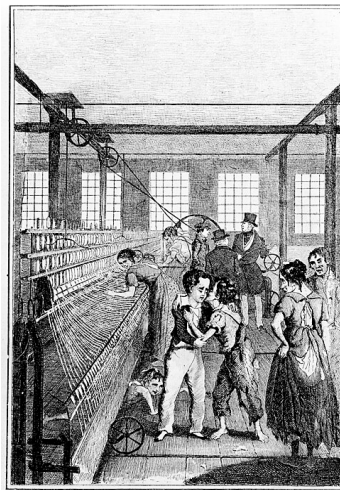
Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Fareham Reds | Pug Mill | Primary Source | Historiography | Factory Acts

Your Sentence:

Source A : Fareham Brickfields



An illustration of child apprentices working in a 19th-century textile factory, highlighting the brutal conditions many children faced.

Before the 1700s, most goods were manufactured by hand in people's homes, known as the 'domestic system'. It is a common myth that this was a relaxed, golden age; in reality, working at home was backbreaking, poorly paid, and children worked long hours weaving or spinning. However, the transition to the 'factory system' completely transformed the scale and danger of British life. As steam engines became more powerful, massive factories were built, causing rapid urbanisation. On a national level, this was incredible progress: Britain became the 'workshop of the world'. But for the workers, the reality was

a brutal shock. The machines now dictated the pace of life. Shifts lasted 12 to 14 hours, six days a week. The noise was deafening, the air was polluted, and heavy machinery lacked safety guards, leading to horrific, unprecedented injuries.

Q1. Change & Continuity: What was the main difference between the child labour used in the 'domestic system' compared to the new 'factory system'?

Factory and mine owners actively sought out child workers. Children were cheap (paid 10% of an adult's wage), easier to discipline, and small enough to crawl under moving machinery. In 1832, Parliament sent Michael Sadler to investigate these conditions. His official report shocked the nation. One textile worker, Matthew Crabtree, was interviewed: 'I began work at eight years old... our common time was from six in the morning till half-past eight at night... we were frequently strapped [beaten] if we were too late or started to fall asleep.' The economic success of the early Industrial Revolution was largely built on the broken backs of these children.

Q2. Source Evaluation: How useful is the 1832 Sadler Report for a historian studying the reality of child labour?

The Industrial Revolution didn't just happen in northern cotton mills; it radically changed the landscape of Hampshire. As cities like London exploded in size, they desperately needed building materials. Fareham sat on a geological goldmine: deep deposits of high-quality clay. During the 19th century, the Funtley Clay Pits and Fareham Brickfields expanded massively. The unique clay produced a brilliant, vibrant red brick famously known as 'Fareham Reds'. These bricks were so highly prized by Victorian architects that they were transported by the newly built railways to construct prestigious buildings in the British Empire, including the Royal Albert Hall in London. Fareham's local mud was literally building the modern world.

Q3. Causation: Sort the following factors into the correct chronological order to explain the boom in Fareham's economy: A) Railways are built to transport heavy goods, B) London's population explodes, C) Fareham Reds are used to build the Royal Albert Hall, D) Massive demand for building materials is created.

While the Fareham Reds brought great wealth to the brickyard owners, the reality for the workers was grueling. In the Funtley brickfields, whole families worked together. Young boys and girls were employed as 'pug boys', driving the horses that turned the pug mills to crush the heavy clay. Others worked as 'carriers', lugging wet clay, or as 'turners', walking up and down long outdoor drying racks to manually flip thousands of heavy bricks so they dried evenly. We know this wasn't just happening in the north of England; if a historian looks at the 1881 Census returns for the Fareham area, they will find numerous local children, some as young as 10, officially listed with occupations like 'brickmaker's labourer' or 'pug boy', proving the local economy relied heavily on child exploitation.

Q4. Source Comparison: Why might a historian need to look at BOTH a national interview (like the Sadler Report) AND a local document (like the 1881 Fareham Census) to fully understand child labour?

As the 19th century progressed, the horrific human cost disgusted the public, leading Parliament to pass 'Factory Acts' to restrict child labour and mandate schooling. But historians still fiercely debate the overall impact of the Industrial Revolution. 'Pessimist' historians (like E.P. Thompson) argue that industrialisation was a catastrophe that punished the working class, destroying their traditional way of life and exploiting

children for decades. Conversely, 'Optimist' historians (like R.M. Hartwell) argue that while the initial transition was harsh, it ultimately represented massive progress; it created the wealth, medical advancements, and eventual higher wages that dramatically increased the life expectancy and standard of living for everyone in the 20th century.

Q5. Extended Writing: Write a structured paragraph answering: 'To what extent was the Industrial Revolution a punishment rather than progress for the working classes?'

L3: Did industrialisation make British towns unlivable? (Urbanisation, Cholera, Public Health; Case Study: Fareham Red Bricks built Portsmouth & London)

LEARNING OBJECTIVES

- ☐ Understand the core themes of Did industrialisation make British towns unlivable? (Urbanisation, Cholera, Public Health; Case Study: Fareham Red Bricks built Portsmouth & London).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: **Urbanisation**

Source A : A Victorian Slum



Source

Overcrowded and dirty streets in a 19th-century British industrial town.

Placeholder narrative.

Q1. Placeholder task.

L4: How was the British Empire built and sustained? (Trade, Raj in India, Naval supremacy; Case Study: Portsmouth Dockyard & Royal Navy)

LEARNING OBJECTIVES

- ☐ Understand the core themes of How was the British Empire built and sustained? (Trade, Raj in India, Naval supremacy; Case Study: Portsmouth Dockyard & Royal Navy).

Do Now Activity

- Recall question from previous lesson.

Vocabulary Check

Write a short paragraph using at least *FOUR* of the vocabulary words below correctly.

British Raj | Naval Supremacy

Source A : Portsmouth Dockyard



Source

HMS Victory and other Royal Navy ships at Portsmouth Dockyard, the heart of British naval power.

Placeholder narrative.

Q1. Placeholder task.

L5: How did ordinary people fight for a voice? (Peterloo, Chartism, Trade Unions; Case Study: Hampshire Swing Riots)

LEARNING OBJECTIVES

- ☐ Understand the core themes of How did ordinary people fight for a voice? (Peterloo, Chartism, Trade Unions; Case Study: Hampshire Swing Riots).

Do Now Activity

- Recall question from previous lesson.

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Chartism | Trade Unions

Your Sentence:

Source A : The Swing Riots



Source

An illustration of agricultural workers protesting against the introduction of threshing machines during the Swing Riots.

Placeholder narrative.

Q1. Placeholder task.

L6: How did the road to democracy expand? (Reform Acts 1832–1884, Secret Ballot; Case Study: Hampshire MPs & Pocket Boroughs)

LEARNING OBJECTIVES

- ☐ Understand the core themes of How did the road to democracy expand? (Reform Acts 1832–1884, Secret Ballot; Case Study: Hampshire MPs & Pocket Boroughs).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: **Reform Acts**

Source A : A Pocket Borough



Source

A political cartoon mocking the corruption of un-reformed electoral constituencies.

Placeholder narrative.

Q1. Placeholder task.

L7: Who party truly benefited from 19th-century transformation? (Synthesis essay balancing economic growth vs. exploitation)

LEARNING OBJECTIVES

- ☐ Understand the core themes of Who party truly benefited from 19th-century transformation? (Synthesis essay balancing economic growth vs. exploitation).

Do Now Activity

1. Recall question from previous lesson.

Vocabulary Check

Write a short paragraph using at least *FOUR* of the vocabulary words below correctly.

Synthesis | Exploitation

Source A : The Two Nations



Source

An illustration showing the stark divide between the rich industrialists and the poor working classes in Victorian Britain.

Placeholder narrative.

Q1. Placeholder task.